Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS, NOV/DEC 2024

DATA STRUCTURES & ALGORITHMS [ELE 4078]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 22 November 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

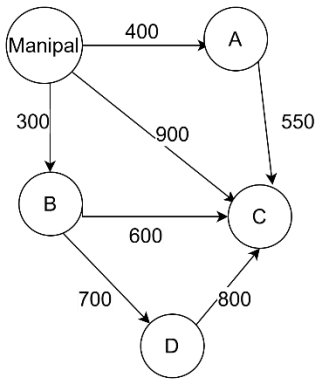
1A. Write a pseudocode algorithm to check whether parentheses are matched in a given string? Use predefined stack functions. ***(02)***

1B. Determine the order in which the multiplications are to be performed with minimum operations using Dynamic Programming.

The order of the matrices are

M1:[2*3],M2:[3*4],M3:[4*5],M4:[5*6] ***(04)***

1C. Analyze the given Graph and obtain the shortest path from Manipal to all other cities (A,B,C,D) in the given graph using Dijkstra's algorithm. Write the pseudo code and show all the steps clearly. The contents of *reached*, *distance,* and *previous* arrays must be shown clearly.



(04)

2A. Obtain post fix expression and evaluate the same using Stack.

6÷2+3-2+5*4+8÷2+3-1*5 ***(02)***

2B. Solve the recurrence relation and determine the time complexity using Substitution method. Write the steps clearly. Assume $2^k=n$

$$T(n) = 2T(n/2) + n \log n \quad \text{where } n>1$$
$$= 1 \quad n=1$$

(04)

2C. It is required to have a change for ₹ 11, and an infinite supply of coins of denominations {1, 2, 5,10} is available. In how many ways can the change be made? Solve the problem using the Dynamic programming technique. Show all the steps clearly. ***(04)***

3A. Construct the binary tree using the following data.
In-order traversal: HFBGIADCEK

Preorder traversal: ABFHGICDEK ***(02)***

3B. Using hash function f(x)= x mod 6, develop the hash table by inserting the following elements: N=6

5,1,80,8,87,52,53,13,34

Use the following collision handling techniques.

i. Open hashing

ii. Linear Probing

iii. Quadratic Probing ***(04)***

3C Deduce the Minimum Spanning Tree (MST) of the graph starting from vertex A using Prim's Algorithm, for the following edge weights. If the starting vertex changes from A to C, does the MST change?

EDGE	WEIGHT
A-B	2
A-C	3
A-D	5
B-D	7
C-D	5
C-E	4
D-E	6

(04)

4A. Implement a queue data structure with enqueue and dequeue operations using pseudocode. ***(02)***

4B. Given a set of characters and their frequencies, use Huffman's Coding algorithm to generate an optimal binary code for each character.

CHARACTER	FREQUENCY
A	5
B	9
C	12
D	13
E	16
F	45

i)Construct the Huffman Tree and generate the codes

ii)Calculate the savings in bits over fixed length encoding

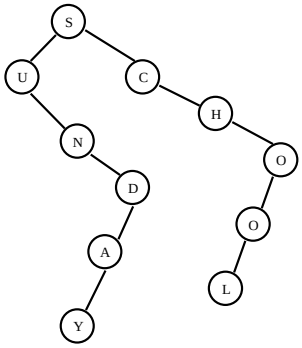
iii)If the frequencies of characters B and C were swapped, would the Huffman codes for any character's change? ***(04)***

4C. Using Kruskal's Algorithm, find the Minimum Spanning Tree (MST) of the graph. Consider the following undirected, weighted graph represented by the edges and their weights: ***(04)***

Edge cost List: (A, B) – 4, (A, C) – 1, (B, C) – 3, (B, D) – 2, (C, D) – 5, (C, E) – 6, (D, E) – 7

If an additional edge (E, A) with weight 8 is added to the graph, does the MST change? Justify your answer.

5A. For the given tree give pre order and in order traversal.



(02)

5B A sorted list of integers is given: [2, 5, 7, 12, 14, 19, 23, 27, 31, 34, 42]. It is required to search for the number 23 on the list. Perform Linear Search technique and describe how many comparisons it would make in the worst case and average case. ***(03)***

5C Given a graph representing 4 cities (A, B, C, D) along with the distances between them as provided in the following table, determine the optimal solution to the Traveling Salesman Problem (TSP) using dynamic programming (DP). Additionally, solve the problem using a greedy algorithm. After obtaining the solutions from both approaches, provide comments on the results and discuss any differences observed.

	A	B	C	D
A	0	10	15	20
B	10	0	35	25
C	15	35	0	30
D	20	25	30	0

(05)